Counters

Laboratory Assignment 10
ECE 201: Digital Circuits and Systems

Learning Objectives

After completing this lab, you will be able to

- design a counter using flip-flops
- describe sequential logic circuits using VHDL
- implement a sequential logic circuit on a FPGA and utilize clocking resources on a FPGA development board.

Equipment and Materials

- DE10-Lite Board
- Intel Quartus Prime Design Software v15.1 or v18.1

Pre-Lab Assignment

Prior to attending your assigned laboratory section, complete the following tasks

- Read section 3.2 of the DE10-Lite User Manual . This section explains how to use the internal clocking circuitry on the DE-series board

Part I: Design of Synchronous Counters Using Behavioral HDL

While we can build counters in VHDL out of individual flip-flop components, another way to specify a counter is using behavioral VHDL and arithmetic, treating the counter as a register and adding 1 to its value:

$$\mathrm{Q(t+1) = Q(t) + 1;}$$

Implement a 4-bit counter that counts up from 0 to 9 (decimal) using **behavioral VHDL (process blocks)** and arithmetic as shown above. When the counter reaches 9 it should return to 0 on the next clock edge and continue counting. Include *Enable* and *Clear* inputs, where the count only advances when *Enable* is 1, and *Clear* restarts the count at 0.
Note: You may get errors if using an output signal on the right side of a signal assignment, so use an intermediate signal to code the addition written above. After running analysis and synthesis, view the RTL Viewer and Technology Map Viewer and include screenshots in your report. Simulate the design to verify its functionality. Then, implement the counter on your DE-series board, using the following pin assignments:

1. Use the pushbutton KEY_0 as the *Clock* input. To successfully use the pushbutton as a clock signal, **use the provided debounce.vhd and instructions to debounce the button signal for the clock input.** Use the pushbutton as the "button" input to the debounce component, use the "result" output as the clock input to the circuit, and connect the debounce.vhd "clk" to PIN_P11 for the 50 MHz system clock.

2. Use switches SW_1 and SW_0 as *Enable* and *Clear* inputs, respectively

3. Use 7-segment display *HEX0* to display the count. Utilize your 7-segment decoder from Lab 7 or the provided hex7seg.vhd component.

Part II: Clocking Circuitry

Follow the steps below to design and implement a circuit that successively flashes digits 0 through 9 on the 7-segment display *HEX0*, displaying each digit for about one second:

1. Design a counter that increments at one-second intervals using the 50-MHz clock signal provided on your DE10 board.

   (a) Figure 1 shows how a large bit-width counter can be used to produce an enable signal for a smaller counter. The rate at which the smaller counter increments can be controlled by choosing an appropriate number of bits in the larger counter.

   (b) **Do not use any other clock signals in your design – make sure that all flip-flops in your circuit are clocked directly by the 50-MHz clock signal.**

   (c) A partial design and pseudocode for an 8 MHz clock is shown in Figure 1 and 2. Use this for reference to design your counter using the 50 MHz clock.

2. Download the design onto your board and confirm that the count from 0 - 9 is displayed on HEX0, changing at one-second intervals.

3. **Demonstrate your design to an instructor or TA for the first part of the lab checkoff.**

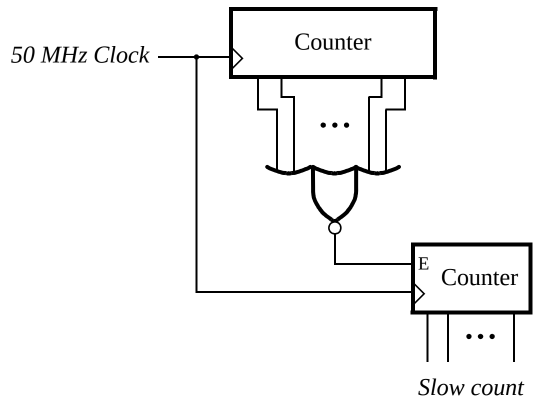

Figure 1: Making a slow counter.

```
--Example slow counter using 8 MHz Clock
process (CLOCK_8)
begin
    if rising_edge(CLOCK_8) then
       --increment fast_count
    end if;
end process;

process (CLOCK_8)
begin
    if (rising_edge(CLOCK_8) and fast_count == 0) then
       --increment slow_count
    end if;
end process;
```

Figure 2: Pseudocode for a slow counter.

Part III: Rotating Word Display

Design and implement a circuit that displays a word with four letters on four 7-segment displays $HEX3-0$. The displays should change at one-second intervals, utilizing your counter from the previous part. There are many ways to design the required circuit – it is recommended that you use behavioral VHDL along with components developed in previous parts of this lab and others.

Count	Characters			
00	d	E	1	0
01	E	1	0	d
10	1	0	d	E
11	0	d	E	1

Table 1: Example rotating the word "dE10" on four displays.

Part IV: Augmenting the Rotating Display

Augment your circuit from the previous part so that it can rotate the word over all of the 7-segment displays on your DE-series board. The shifting pattern for the DE10-Lite is shown in Table 2. **Demonstrate the rotating displays on your DE10 board for the second part of the lab checkoff.**

Count	Character pattern					
000			d	E	1	0
001		d	E	1	0	
010	d	E	1	0		
011	E	1	0			d
100	1	0			d	E
101	0			d	E	1

Table 2: Rotating the word dE10 on six displays.